Lenin T

Coimbatore, Tamil Nadu

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Summary

Electronics and Communication Engineering graduate with hands-on project experience in Python-based backend and full-stack development, REST APIs, SQL databases, and embedded systems. Familiar with data analysis and machine learning fundamentals, with ongoing learning in deep learning and generative AI. Seeking entry-level opportunities across software development, Python/backend development, IT, and AI/ML.

Education

Kumaraguru College of Technology **Nov. 2022 – May 2026**
Bachelor of Engineering in Electronics and Communication – CGPA: 7.9 *Coimbatore, Tamil Nadu*

Sri Vivekananda Matriculation Higher Secondary School
Class XII – 86.6% — *Class X – 96.2%* *Coimbatore, Tamil Nadu*

Skills

Programming: Python; SQL; C; Embedded C; Object-Oriented Programming; Data Structures and Algorithms

Software Development: FastAPI; REST APIs; React; PostgreSQL; SQLite

AI/ML and Data Science: NumPy; Pandas; Data Preprocessing; Exploratory Data Analysis; Statistics; Scikit-learn; Machine Learning; Deep Learning Fundamentals; PyTorch; Generative AI; LLM Fundamentals; Prompt Engineering

Tools and Platforms: Git; GitHub; Docker; Linux; VS Code; Arduino IDE; MATLAB/Simulink

Core Competencies: Problem Solving; Analytical Thinking; Adaptability

Projects

ToLet Coimbatore – Hyperlocal Rental Platform **April 2026 – June 2026**
Tech Stack: Python, FastAPI, PostgreSQL, React, Tailwind CSS, Leaflet.js, Cloudinary, Railway *Coimbatore, Tamil Nadu*

- Built a full-stack hyperlocal room rental platform connecting landlords and tenants in Coimbatore without brokers.
- Developed REST APIs with FastAPI and PostgreSQL supporting listing CRUD, JWT authentication, and photo uploads via Cloudinary.
- Implemented React frontend with Leaflet.js map view, live search, PWA support, and mobile-responsive UI.
- Deployed backend and frontend independently on Railway with environment-based configuration and PostgreSQL cloud database.

Vehicle Platooning for Heavy Duty Vehicles **Dec. 2025 – April 2026**
Tech Stack: ESP32, Arduino UNO, ESP-NOW / RF Communication, Ultrasonic Sensor *Coimbatore, Tamil Nadu*

- Developed a leader-follower vehicle communication prototype using ESP-NOW for real-time coordination.
- Integrated ultrasonic-based obstacle detection with an automatic braking mechanism.
- Demonstrated synchronized vehicle movement for improved transportation efficiency and safety.

Predictive Maintenance in Factories using Embedded System **Feb. 2025 – April 2025**
Tech Stack: ESP32, Ultrasonic Sensor, Gas Sensor, DHT11, Blynk IoT Platform *Coimbatore, Tamil Nadu*

- Developed a cloud-connected industrial monitoring system using ESP32 and multiple environmental sensors.
- Implemented real-time detection of gas leakage, temperature variations, and obstacle proximity with mobile alerts.
- Enhanced preventive maintenance through early fault detection and automated safety response mechanisms.

Certifications

- Introduction to Artificial Intelligence (AI) – IBM, Coursera
- Introduction to Internet of Things – NPTEL (IIT Kharagpur) – Elite (91%)
- Software Defined Vehicles (SDV 101) – Coursera